



WEARABLE SYSTEM FOR MONITORING POSTURE AND HUMAN MOVEMENTS

MS3 16/12/2025



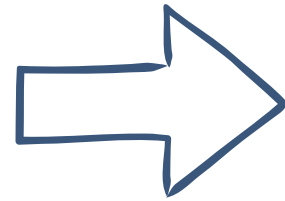
Beatriz Francisco N° 118638
Catarina Ribeiro N° 119467
Mariana Marques N°118971
Marta Cruz N° 119572
Matilde Rodrigues N°119714

MOTION SUIT

Our project involves developing a suit equipped with sensors that measure body posture and detect anomalies



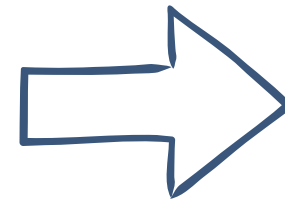
USER STORIES



As a
Suit User

I want
View statistics on
dashboard

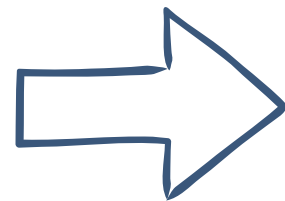
So that
I can see my posture data
over time



As a
Researcher

I want
Access all collected
information

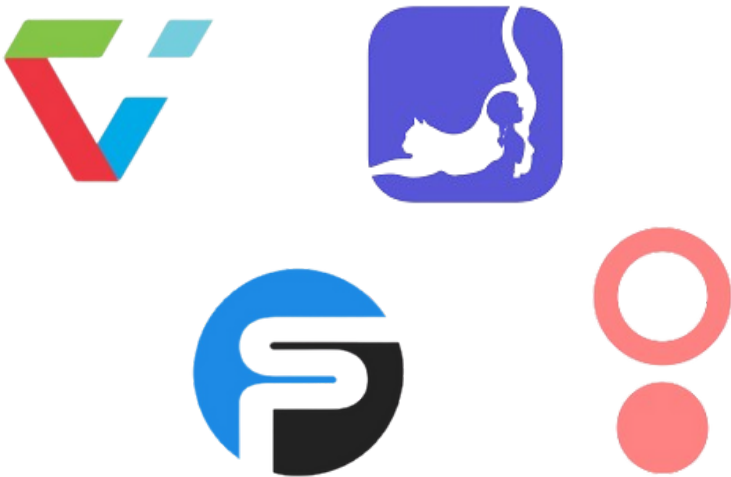
So that
I can identify postural
deviations and abnormal
patterns



As a
Administrator

I want
To manage user
accounts

So that
I can control access and
provide support



RELATED WORK

	SmartPosture	FlexiTrace	SwordHealth	DorsaVI	Current Suit	MotionSuit
Performance Analytics						
Real-Time Feedback						
Wearable Sensors						
Gamification						
3D Model						

CALENDAR

		December	
20/NOV/25	✓ ▼	T-2 View statistics on the dashboard	15/DEC/25
20/NOV/25	✓	Frontend (interface development)	15/DEC/25
20/NOV/25	✓	Database and Frontend Connection	15/DEC/25
20/NOV/25	✓ ▼	T-10 Manage user accounts	15/DEC/25
20/NOV/25	✓	Frontend (interface development)	15/DEC/25
20/NOV/25	✓	Database and Frontend Connection	15/DEC/25
20/NOV/25	✓ ▼	T-9 Access all collected information	15/DEC/25
20/NOV/25	✓	Frontend (interface development)	15/DEC/25
20/NOV/25	✓	Database and Frontend Connection	15/DEC/25
20/NOV/25	✓	Milestone 3	16/DEC/25

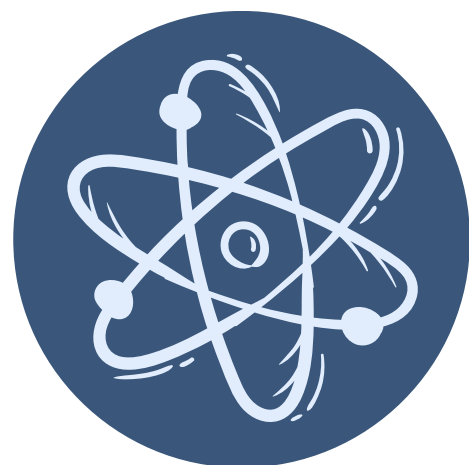
CHALLENGES



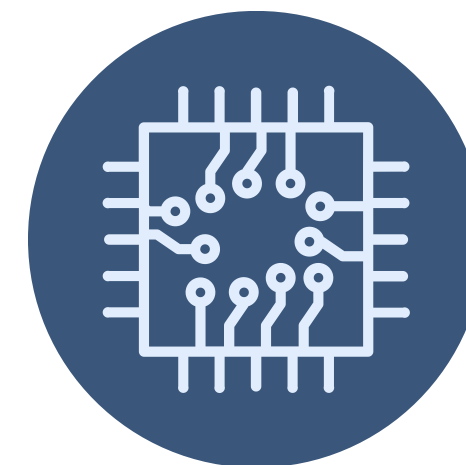
Complexity in managing different types of **data**



Real-time **communication** between the Raspberry Pi and the backend

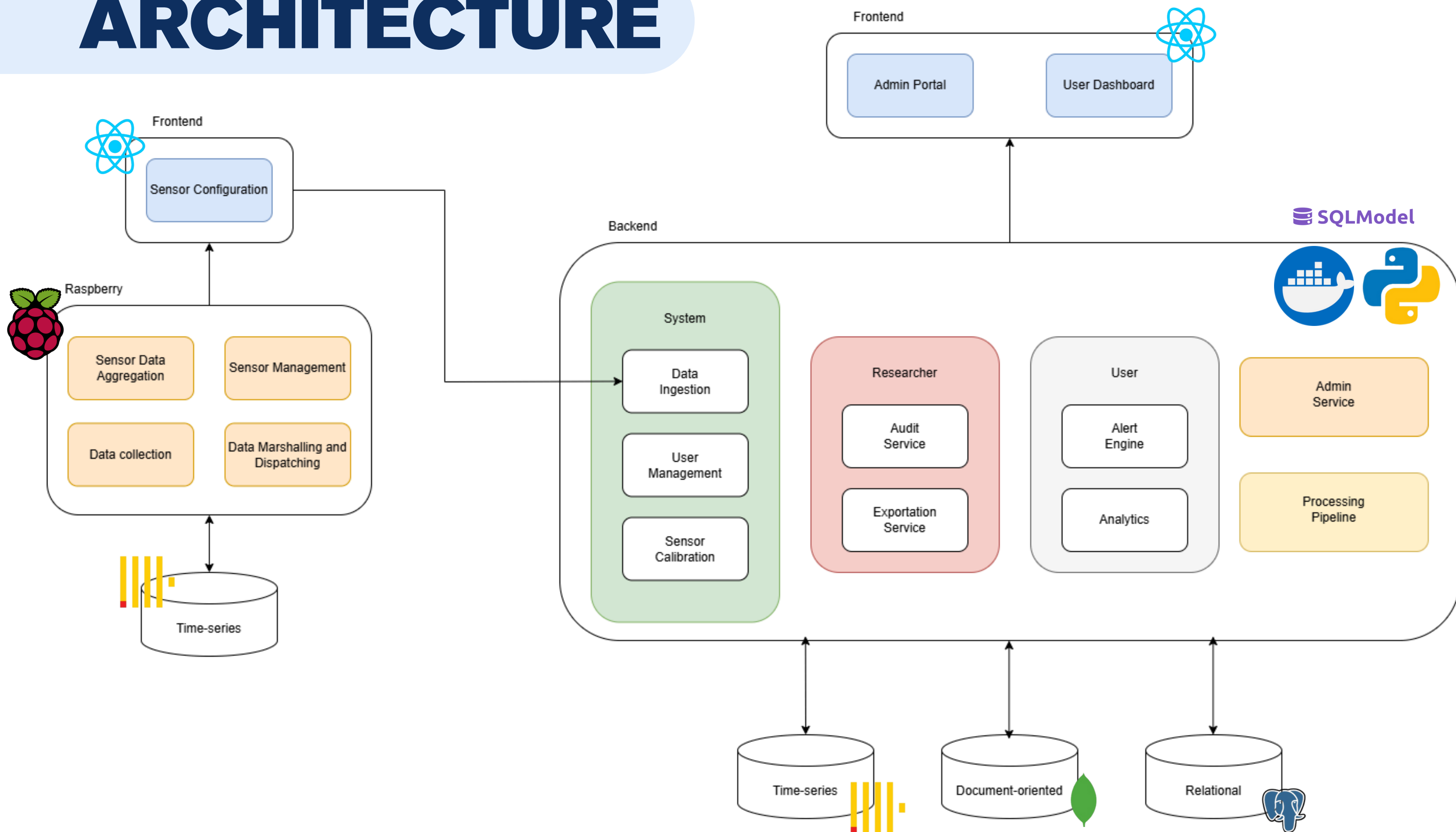


Working with **physics** in data generation simulation



Sensor configuration

ARCHITECTURE

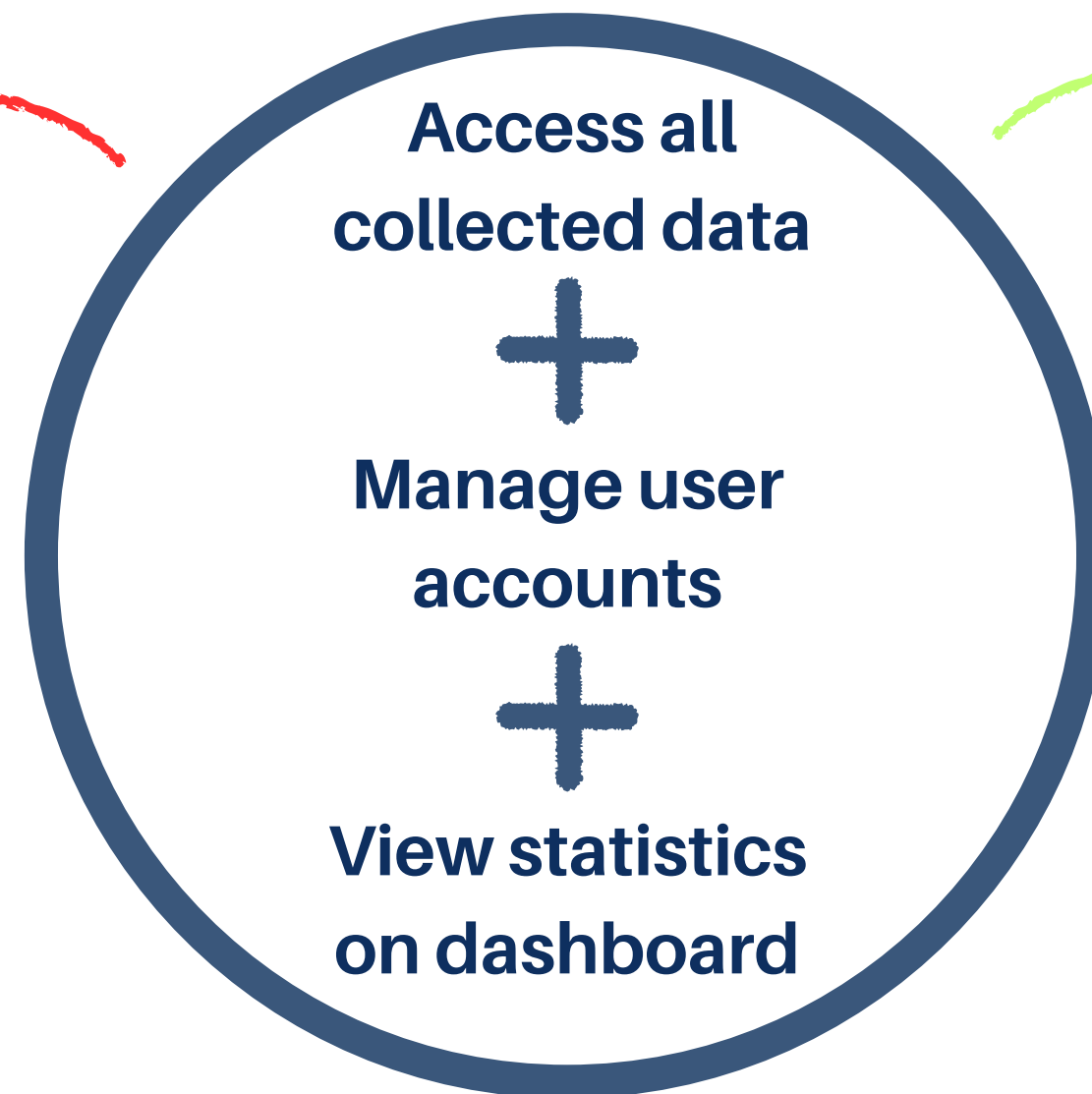


TASKS

- **Incompatible** data types
- A single database would make the system **slow** and **difficult to scale** to "Timeseries".



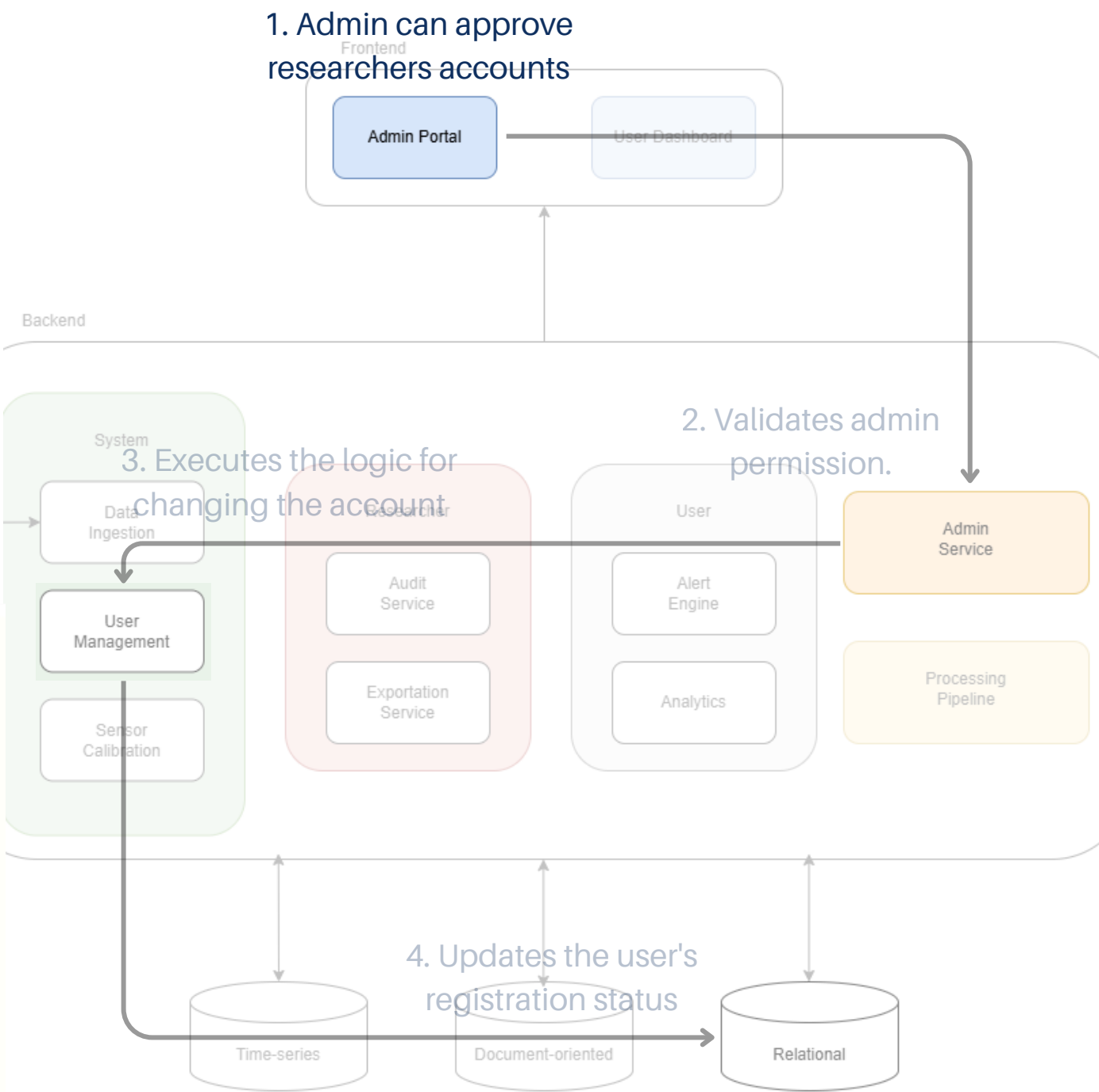
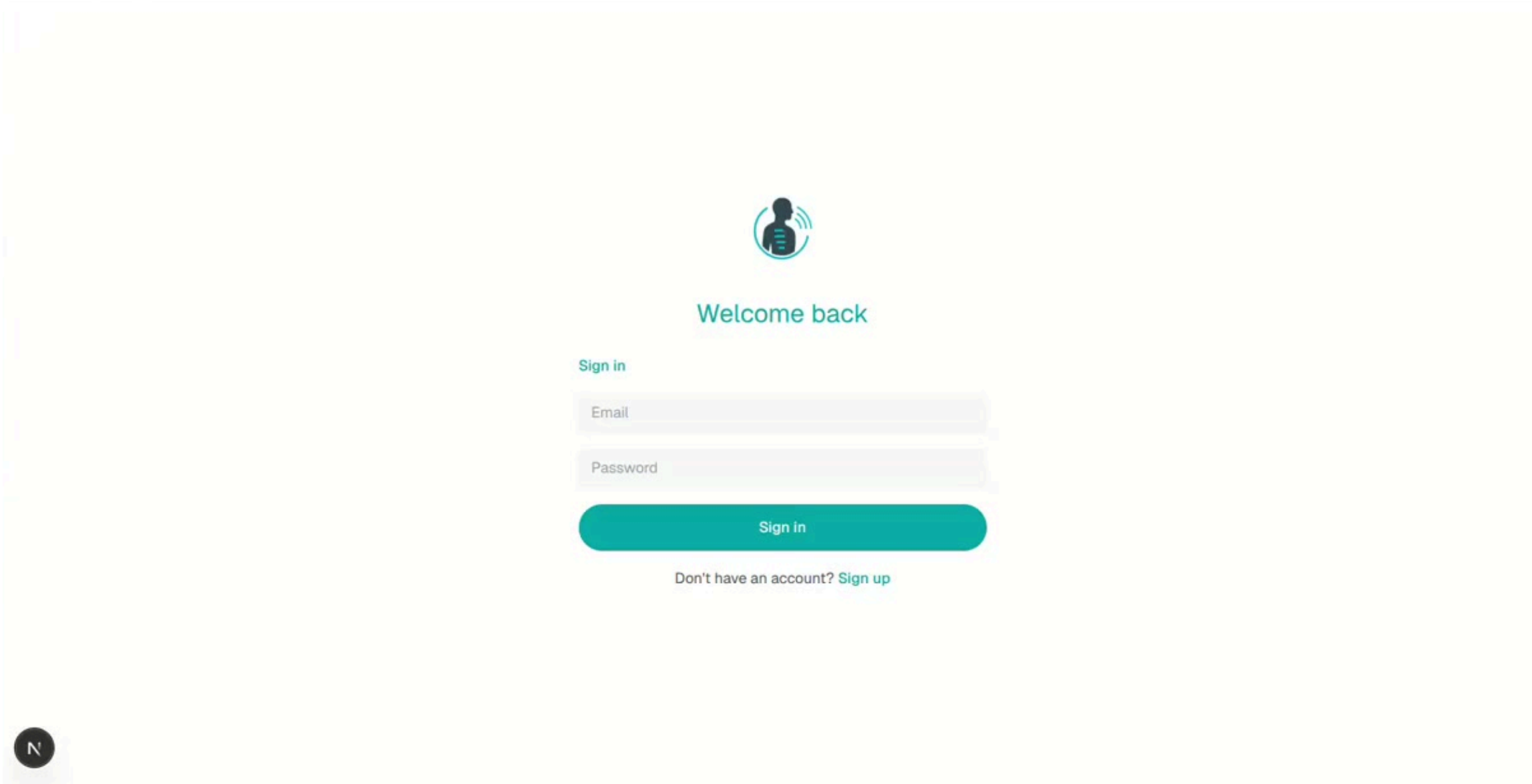
Complexity in managing different types of **data**



- Creation of **three** databases (backend)
- **Export** data
- Better user **management**

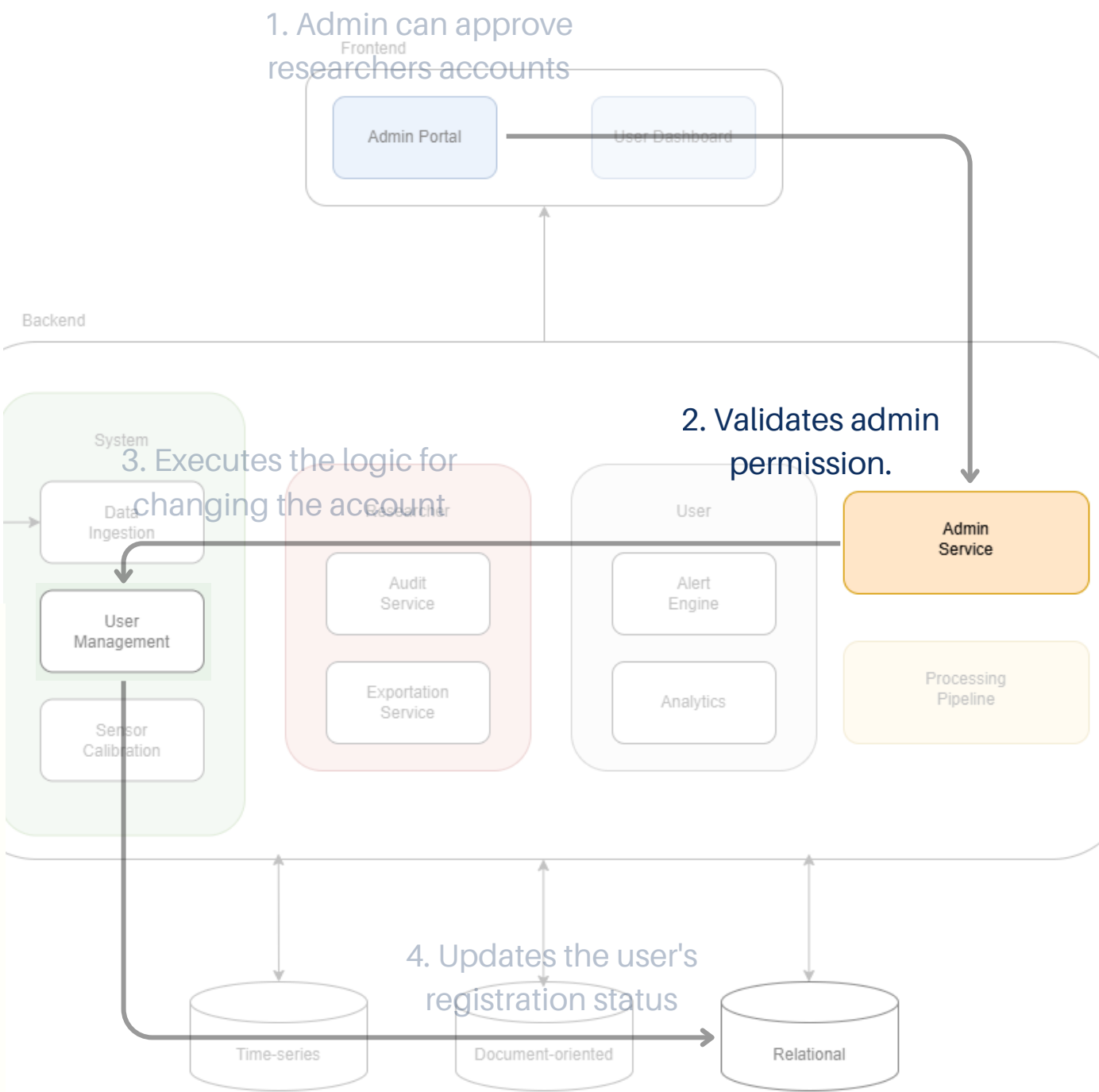
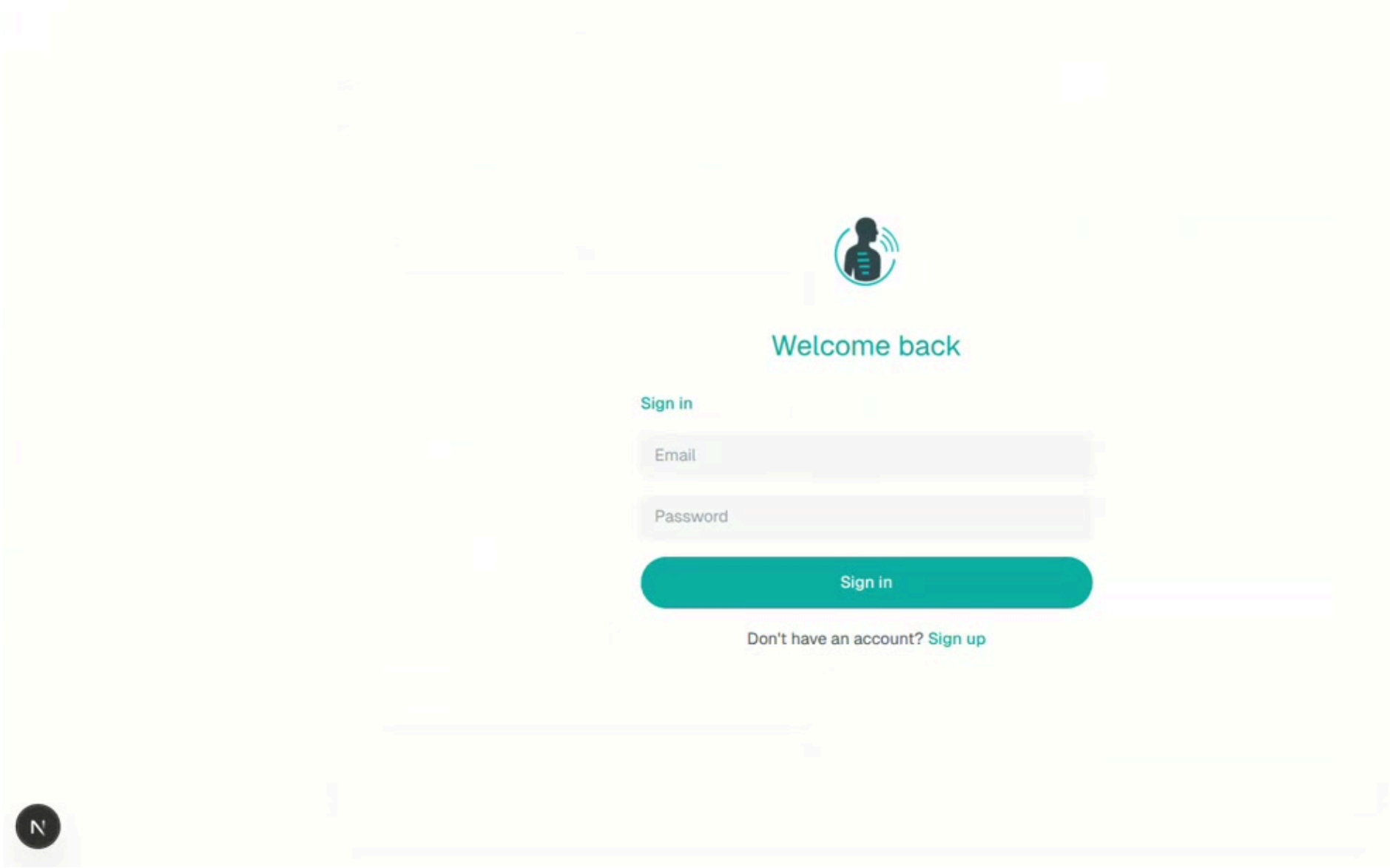
WORKFLOW

Manage user accounts



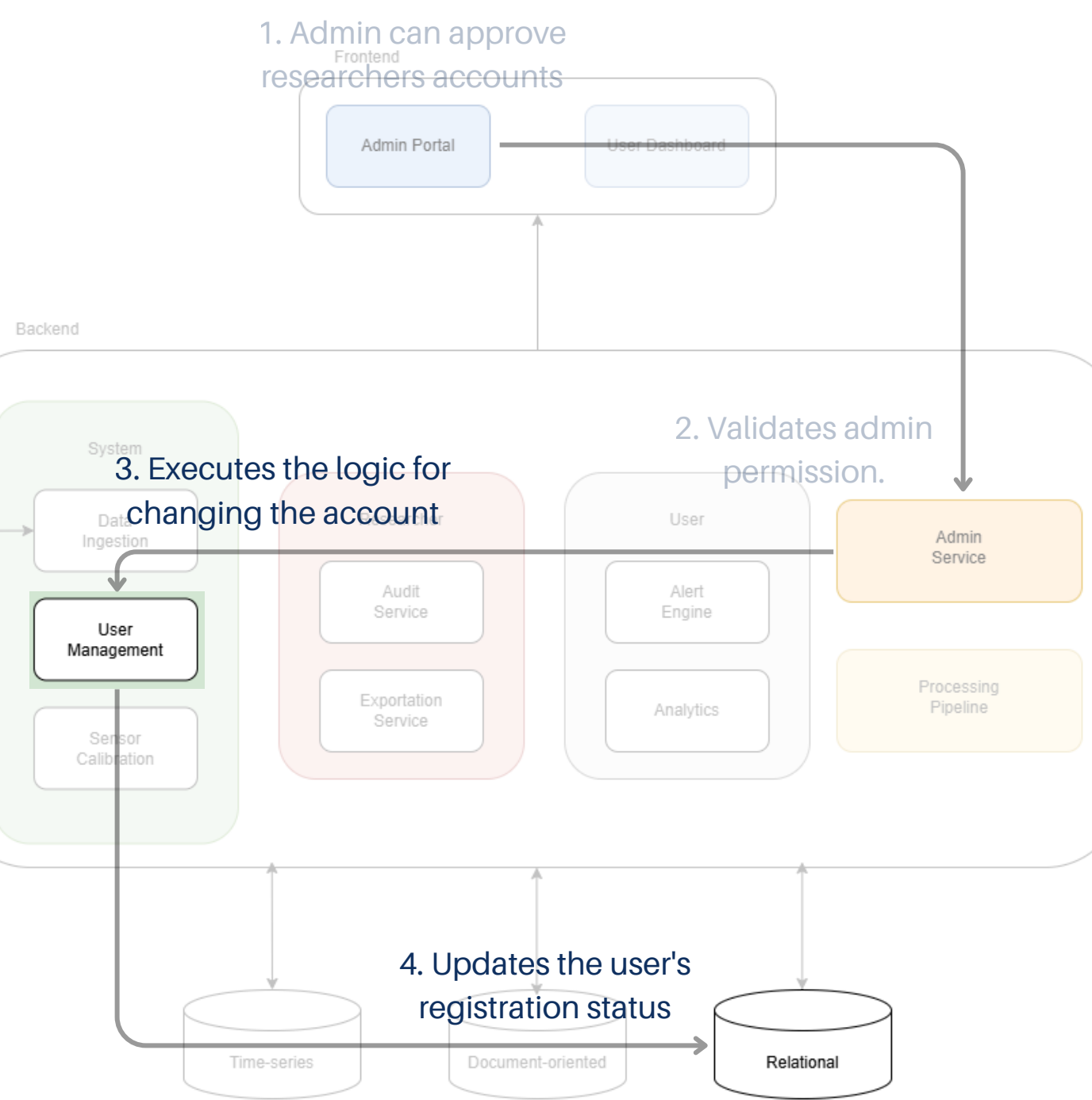
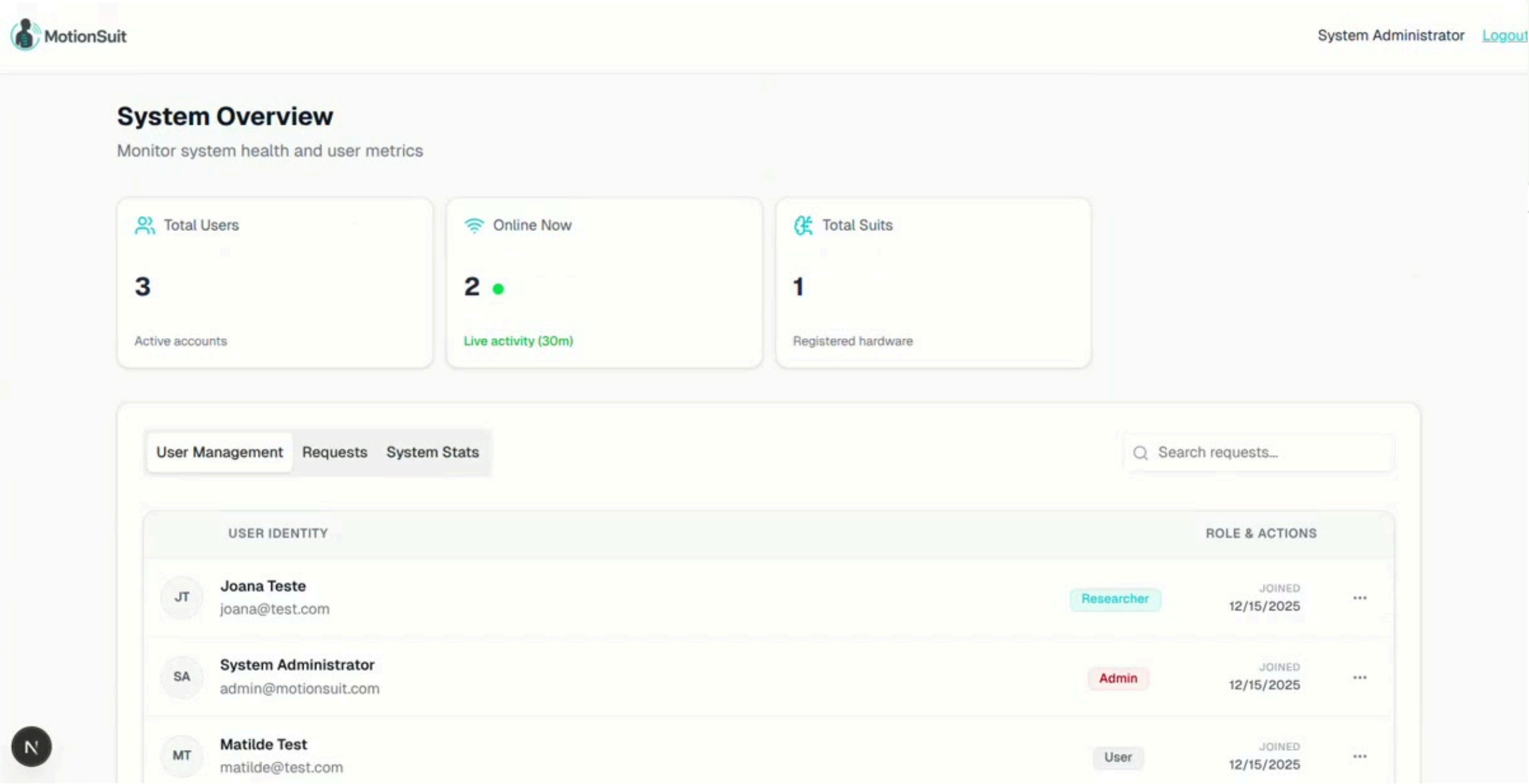
WORKFLOW

Manage user accounts



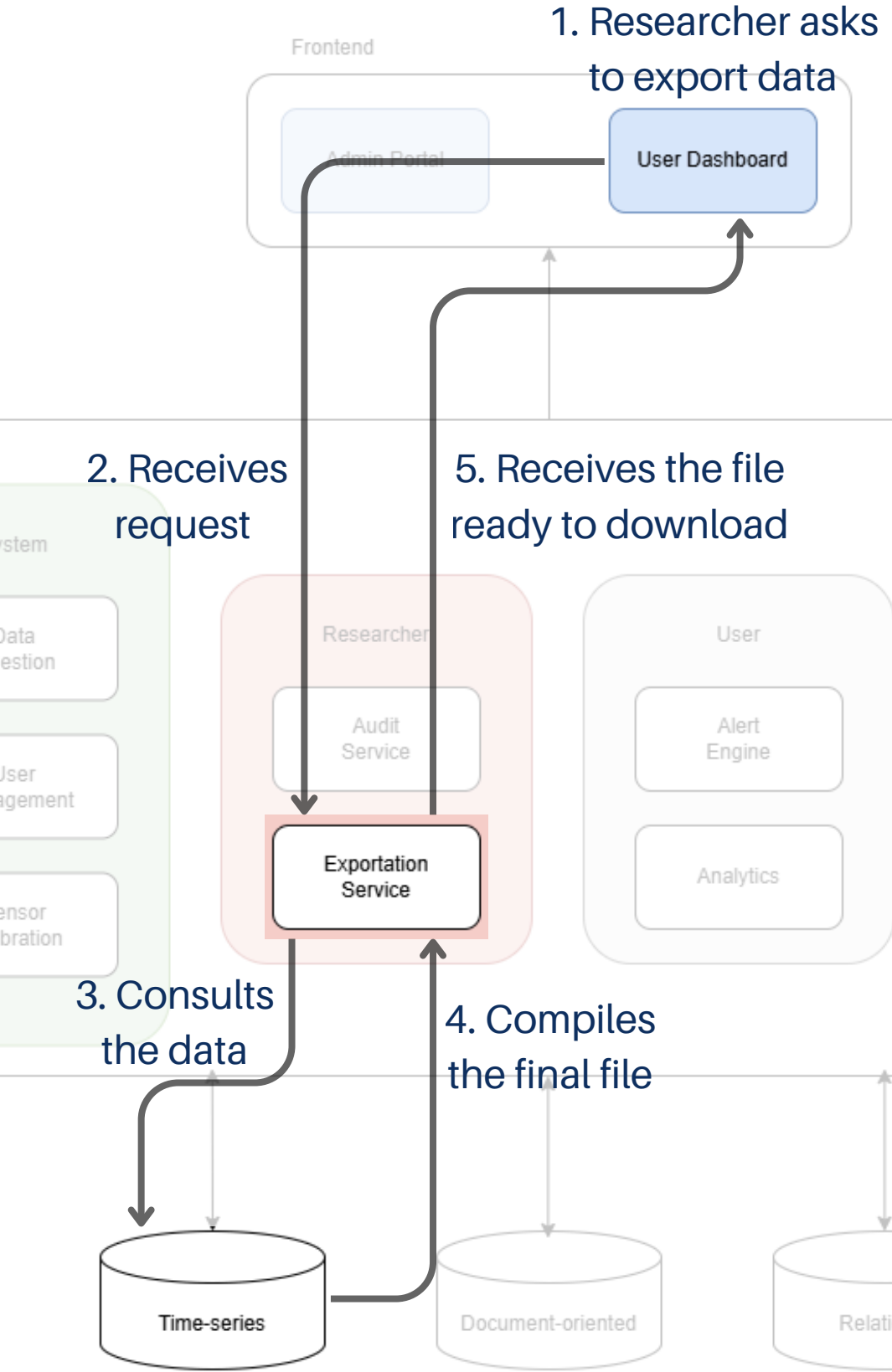
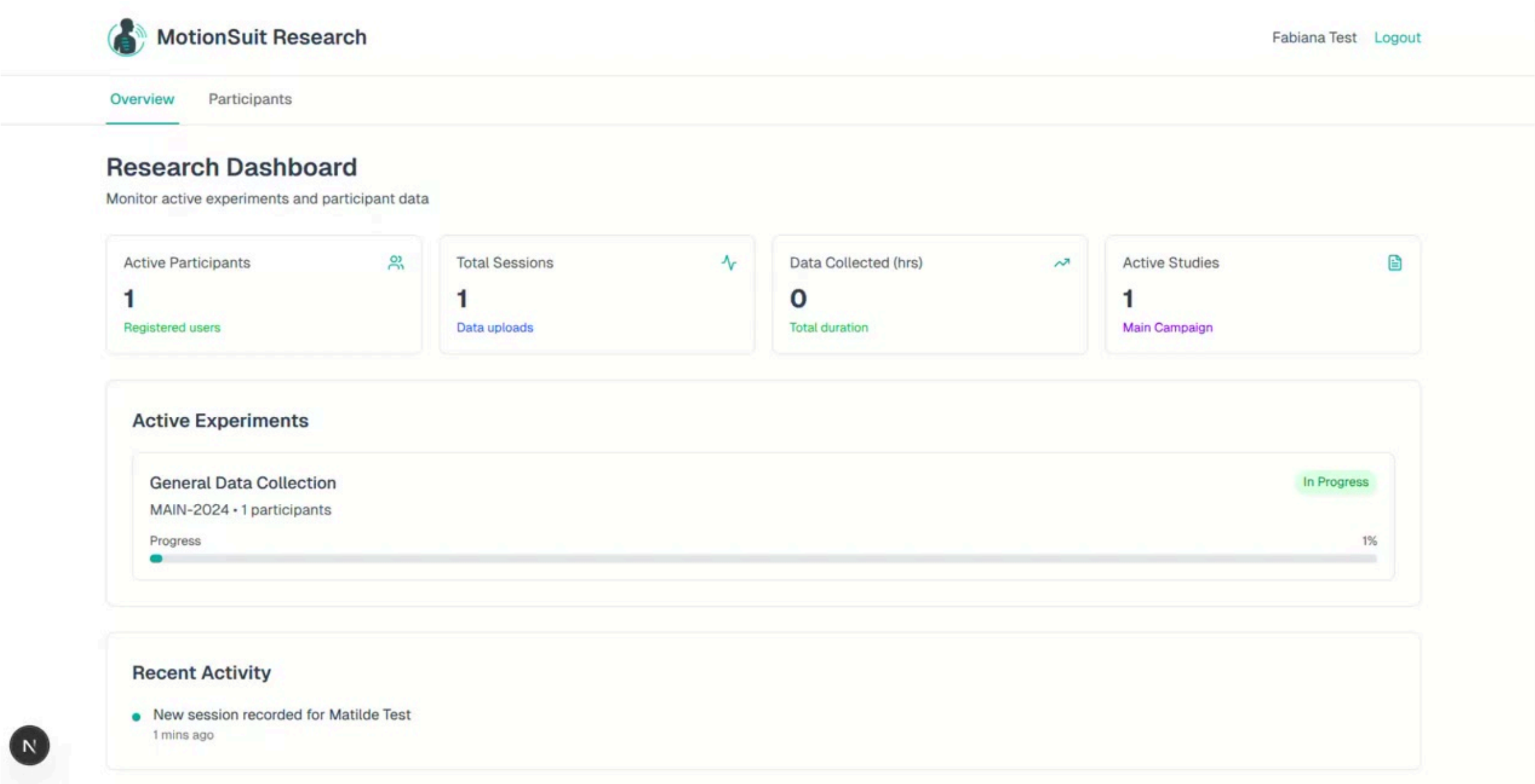
WORKFLOW

Manage user accounts



WORKFLOW

Access all collected data



WORKFLOW

View statistics on dashboard



Profile

**Marta Cruz**
Suit User

Email

cruz@ua.pt

Role

Suit User

Birth Date

January 1, 1990

Gender

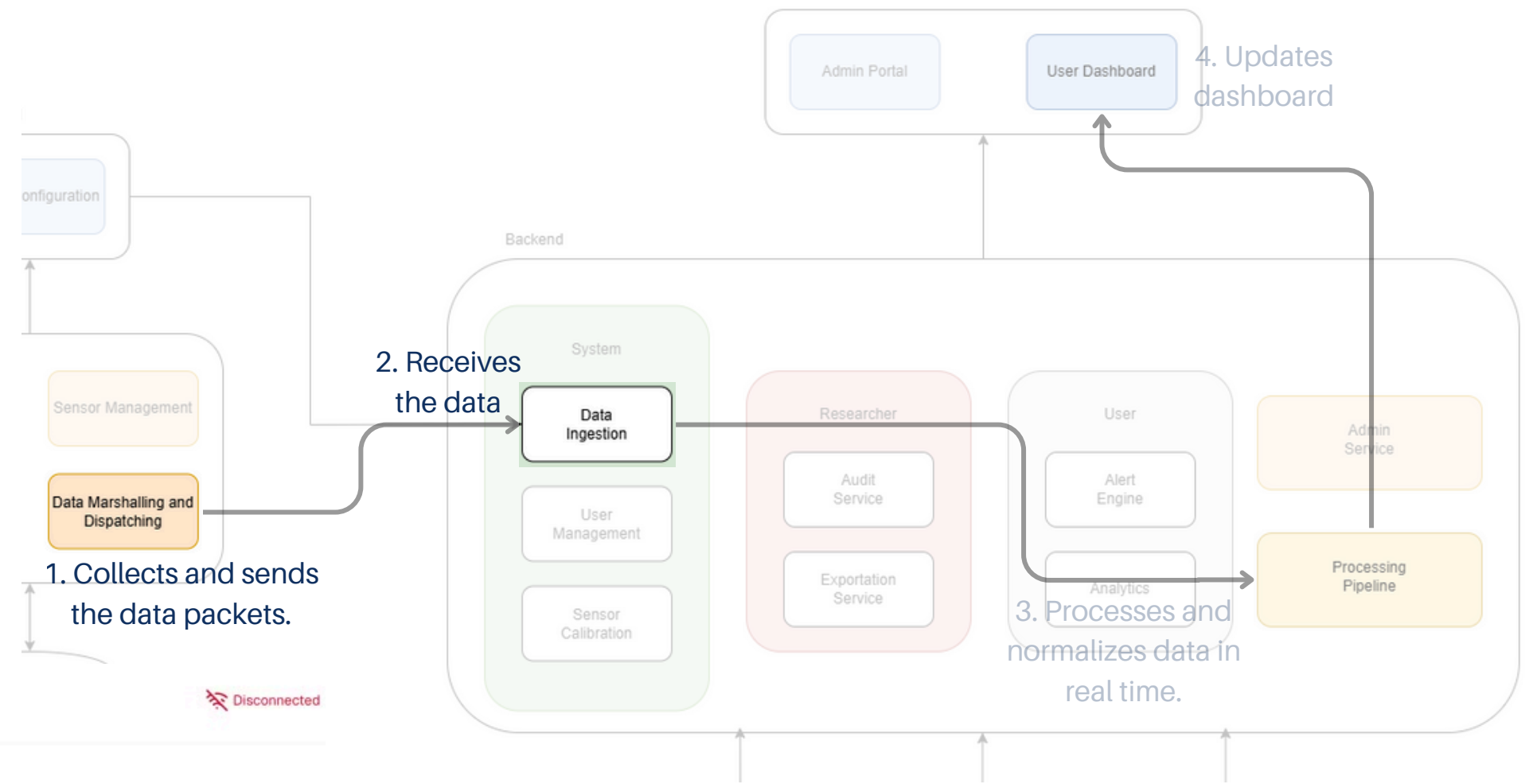
Female

MotionSuit Connection

Disconnected

No suit connected to your account

Connect Suit



WORKFLOW

View statistics on dashboard



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Device ID

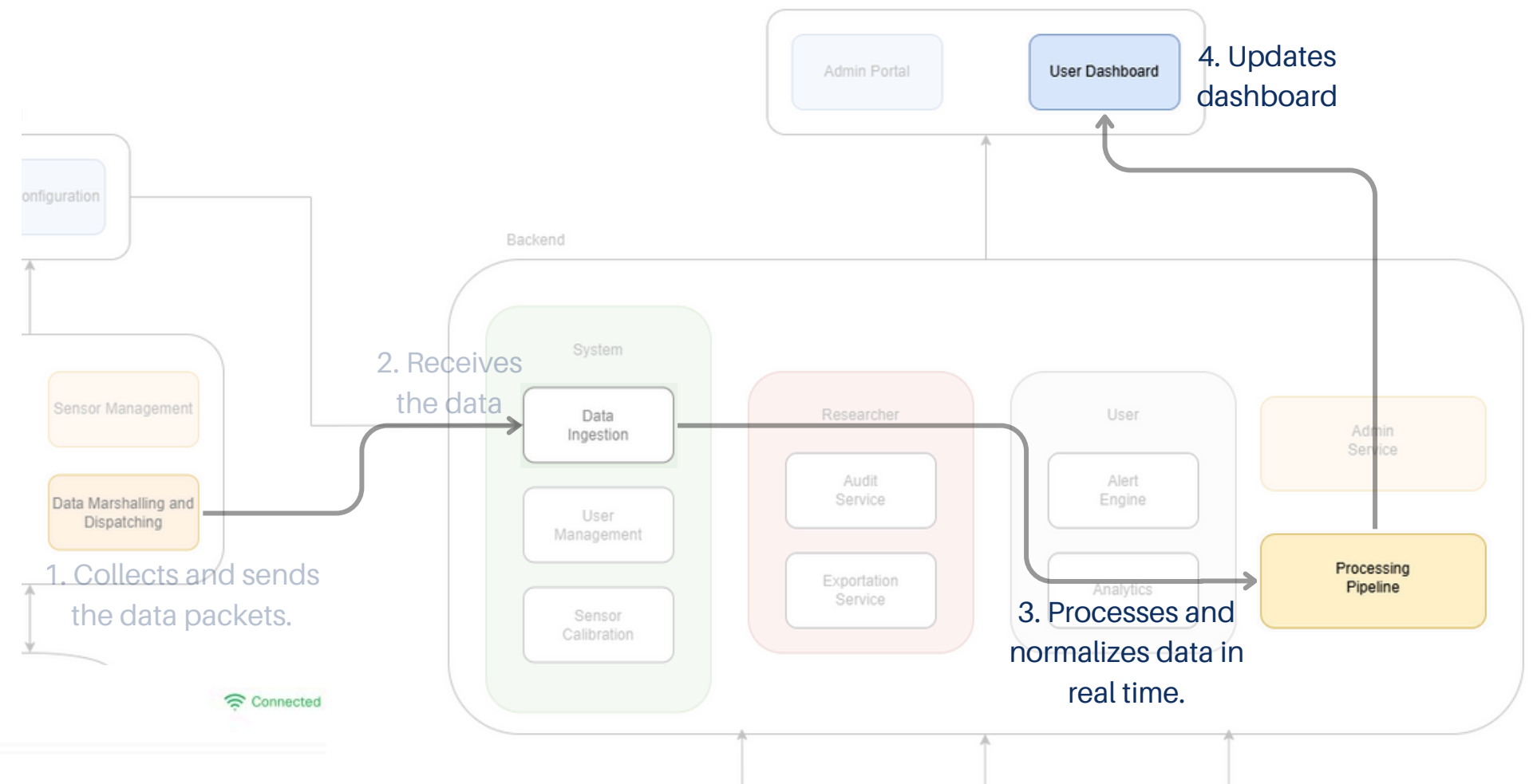
MS-2025-123456

Disconnect

Change Device

Session History (12)

Date	Time	Duration	Frames	Export
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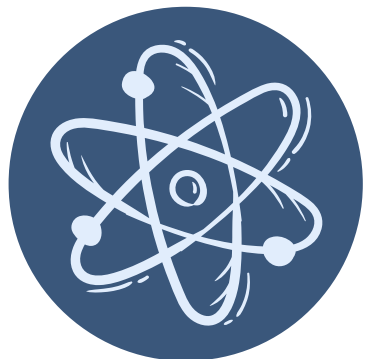


TASKS

- Generate **random values** would cause the 3D model to assume anatomically impossible positions
- Complexity of **simulating physics**

Frontend (interface development)
+
View statistics on dashboard

- Physics-based **data generation** script
- Visualization of simulated data on a **3D model** in the dashboard



Working with **physics** in data generation simulation

WORKFLOW

View statistics on dashboard



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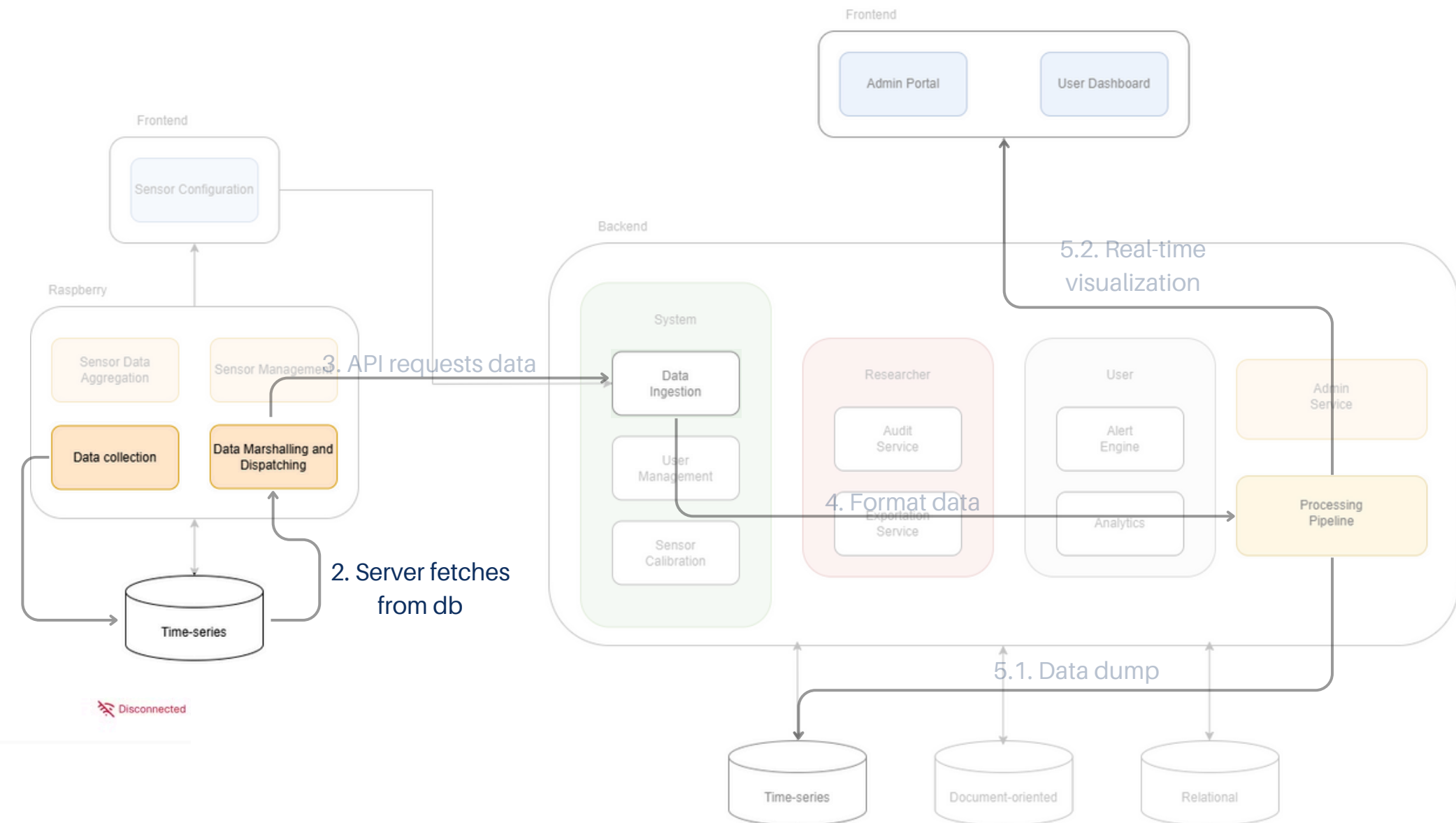
MotionSuit Connection

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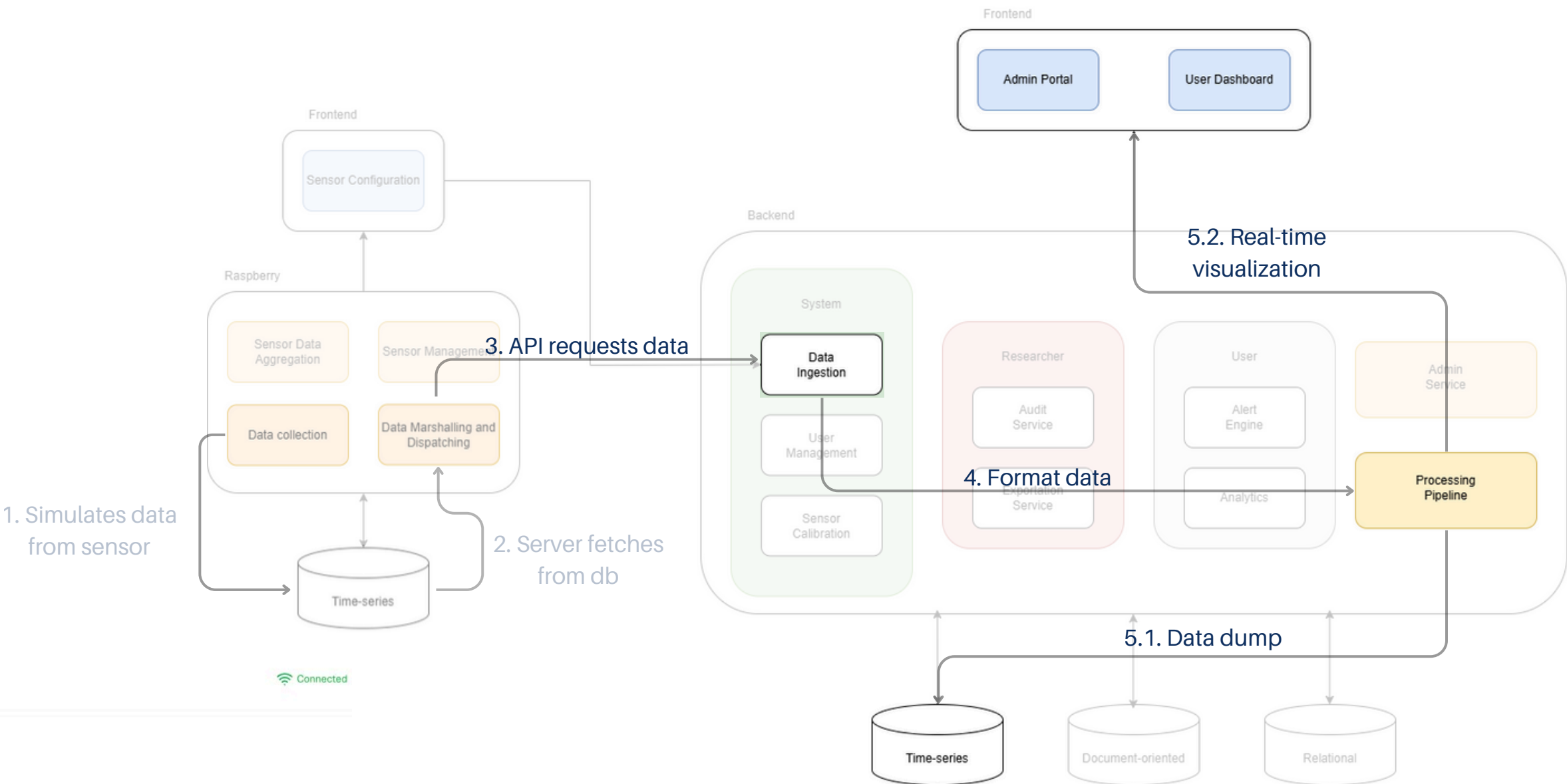
Connect Suit

1. Simulates data from sensor



WORKFLOW

View statistics on dashboard



MotionSuit

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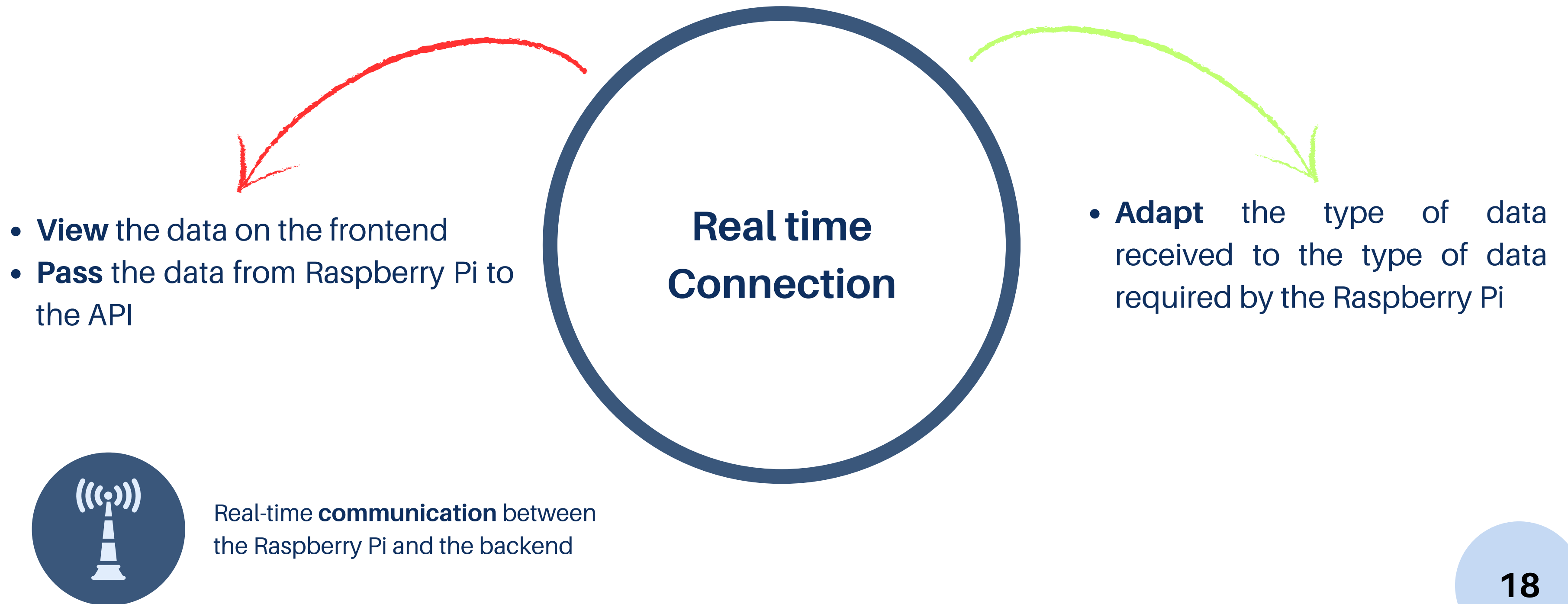
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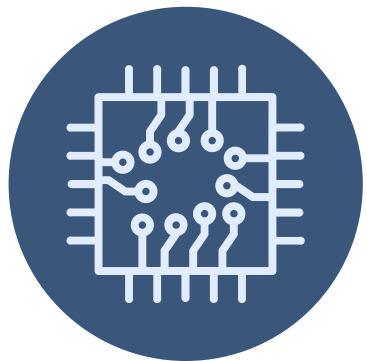
TASKS



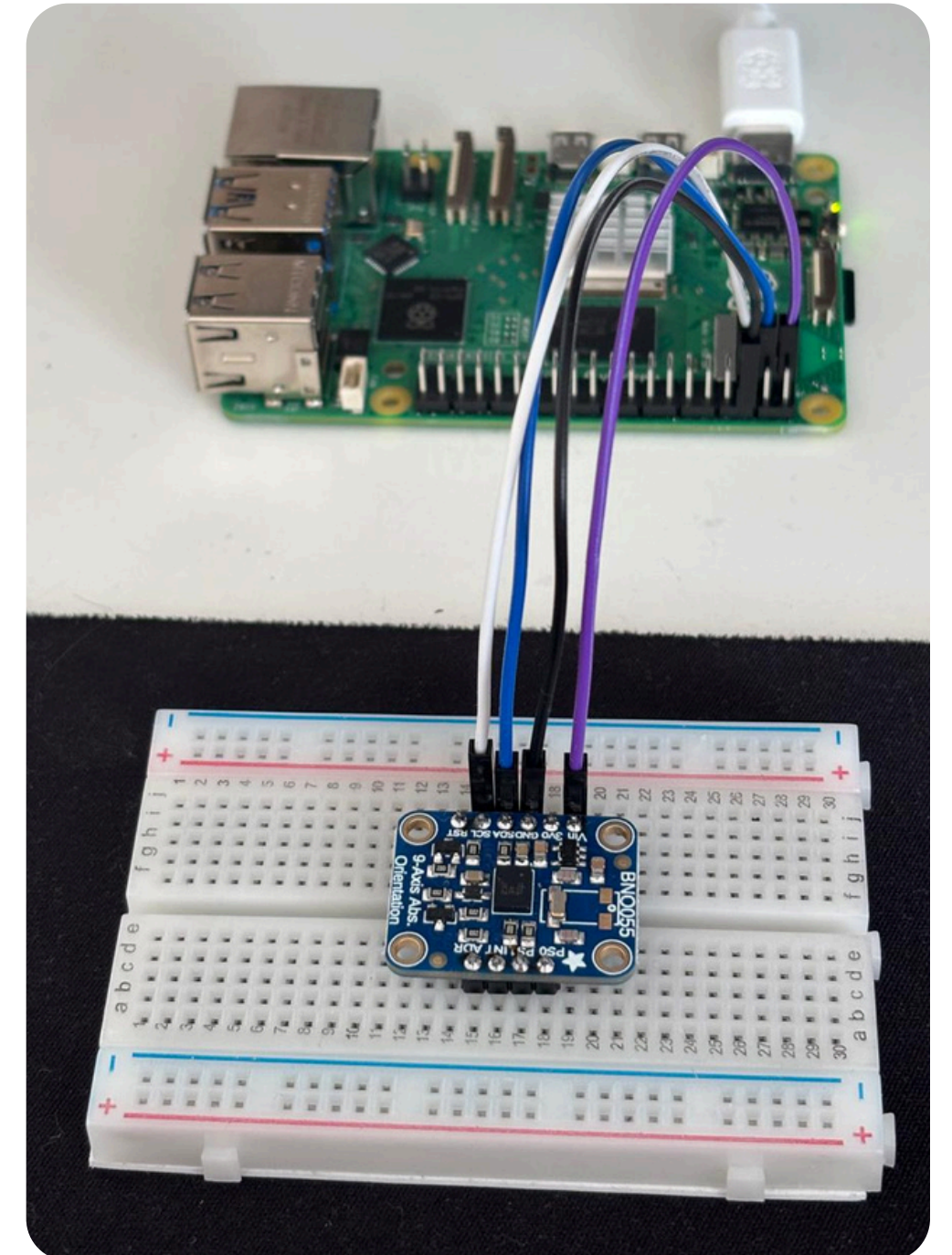
SENSOR

- Never worked with sensors
- Difficulty in **sending the data** retrieved from the sensor to the Raspberry Pi

Sensor Configuration

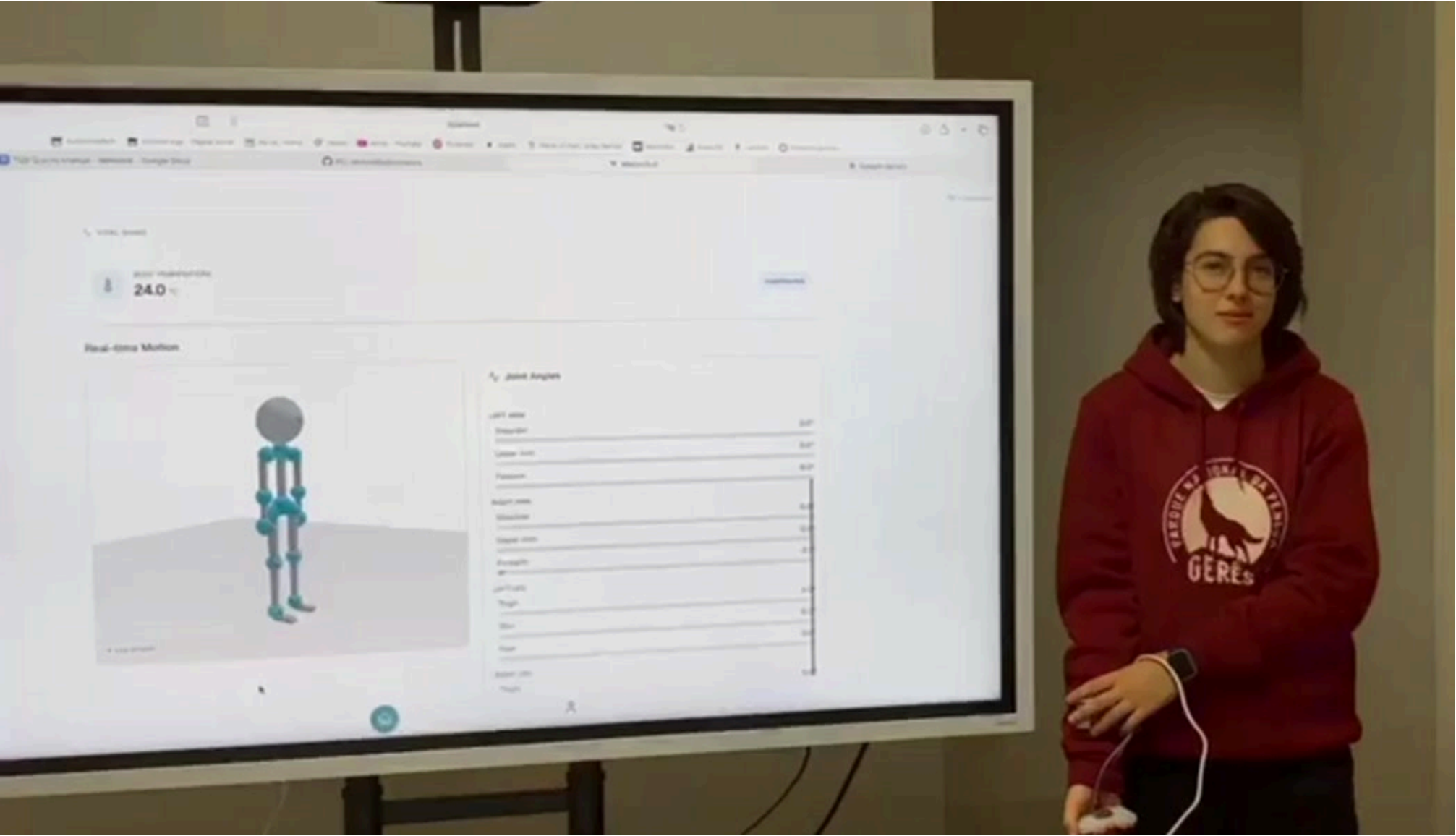
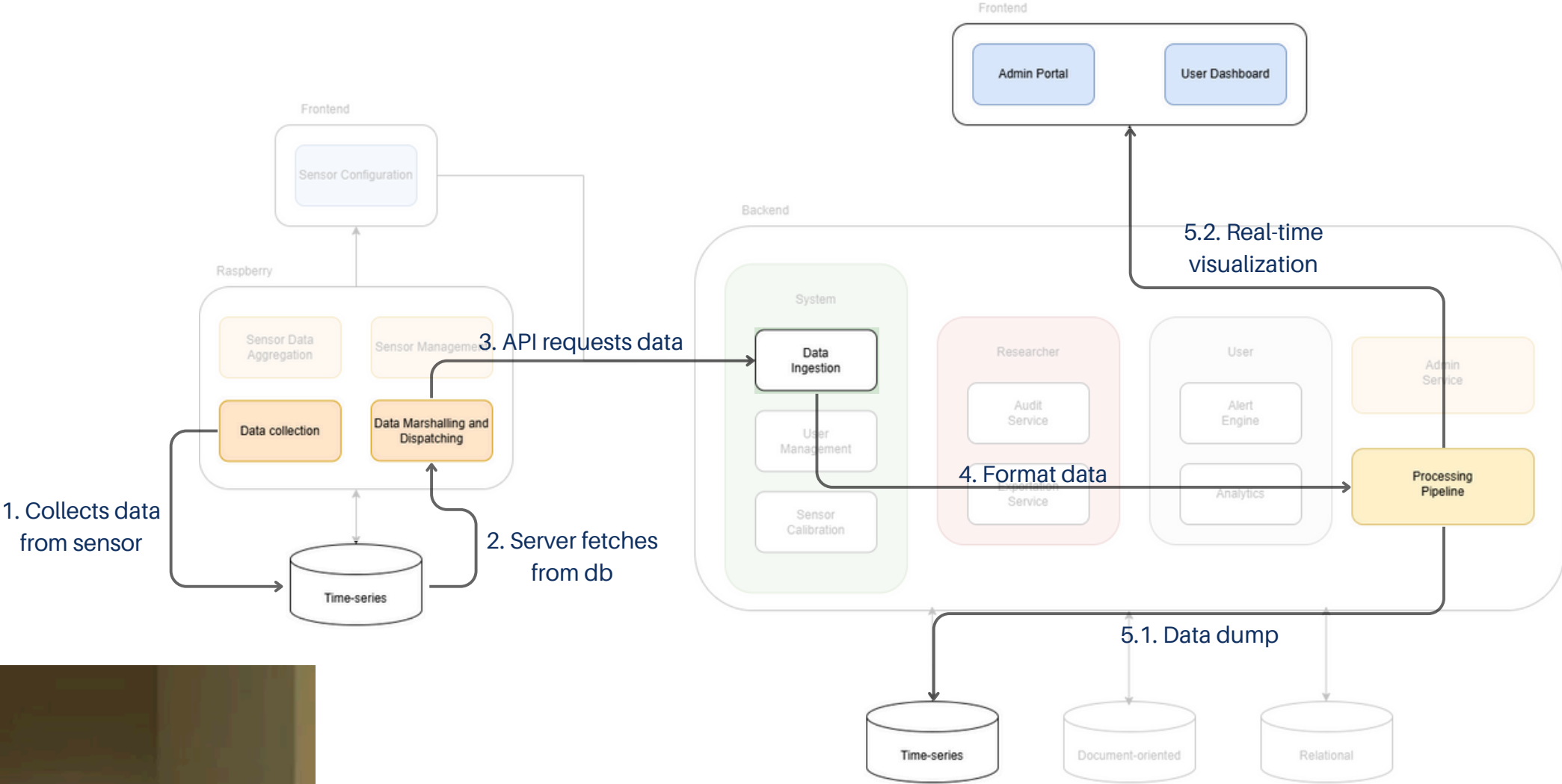


Sensor configuration



SENSOR

Real sensor



MAIN FEATURES

Suit User	Researcher	Admin
See posture in 3D model	Filter data with various attributes	Access to all users
Export data from a specific session or all sessions, in different formats	Export data from a specific user, in different formats	Give researcher permission
See joint angles		Can change user's passwords

WORK

Defined Work	Extra	Future Work
Frontend with backend connection	Sensor sending real data	Performance analysis
Simulated data and access to all of it		14 Sensors
Functional dashboard		Gamification
User management		



<https://pei-motionsuit.github.io/microsite/>

THANK YOU!



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